

Tao Wen

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Research Interests

Generalizable, Reliable, and Trustworthy Machine Learning and AI

Education

- Sep. 2024 – present  **Dartmouth College**, Hanover, NH, U.S.
Ph.D. student in Computer Science, Advisor: Yujun Yan
- Sep. 2022 – May 2024  **New York University**, New York, NY, U.S.
Master of Science in Data Science, Center for Data Science
- Sep. 2018 – Jun. 2022  **Renmin University of China**, Beijing, China
Bachelor of Science in Data Science, School of Statistics

Research Publications

- 1 **T. Wen**, E. Chen, Y. Chen, and Q. Lei, “Bridging domain adaptation and graph neural networks: A tensor-based framework for effective label propagation,” in *Conference on Parsimony and Learning*, B. Chen, S. Liu, M. Pilanci, *et al.*, Eds., ser. Proceedings of Machine Learning Research, vol. 280, PMLR, 24–27 Mar 2025, pp. 599–614. 🔗 URL: <https://proceedings.mlr.press/v280/wen25a.html>.
- 2 **T. Wen**, Z. Wang, Q. Zhang, and Q. Lei, “Elastic representation: Mitigating spurious correlations for group robustness,” in *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*, Y. Li, S. Mandt, S. Agrawal, and E. Khan, Eds., ser. Proceedings of Machine Learning Research, vol. 258, PMLR, Mar. 2025, pp. 541–549. 🔗 URL: <https://proceedings.mlr.press/v258/wen25a.html>.
- 3 **T. Wen**, E. Chen, and Y. Chen, “Tensor-view topological graph neural network,” in *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, S. Dasgupta, S. Mandt, and Y. Li, Eds., ser. Proceedings of Machine Learning Research, vol. 238, PMLR, Feb. 2024, pp. 4330–4338. 🔗 URL: <https://proceedings.mlr.press/v238/wen24a.html>.

Experiences

Research

- Sep. 2025 –  **Large-scale AI Model Diagnosis**
Working to develop a scalable model diagnostic framework to assess the generalization of scientific machine learning and large language models. Work in progress. Funded by DARPA as a graduate research assistant.
- Sep. 2024 – Jan. 2026  **Diagnosing Graph Neural Networks via Feature Learning**
Proposed the first practical and systematic framework to diagnose the generalization of graph neural networks via feature learning. Work accepted by GFM workshop at ICML’26 and under review at NeurIPS’26.
- May 2024 – Oct. 2024  **Mitigating Spurious Correlations for Group Robustness**
Proposed Elastic Representation to address spurious correlations and improve worst-group accuracy without a sacrifice of overall accuracy. Work published at AISTATS 2025.

Experiences (continued)

- Aug. 2023 – Jan. 2024  **Domain Adaptive Graph Classification via Label Propagation**
Developed an approach that bridged label propagation, domain adaptation, and graphs to improve graph classification performance on data from new domains. Work published at CPAL 2025.
- Mar. 2023 – Aug. 2023  **Tensor-view Topological Graph Neural Network**
Developed the first approach bridging tensor methods with multi-modal graph topological features to improve graph neural network generalization. Work published at AISTATS 2024.

Teaching

- Jan. 2025 – Mar. 2025  **Teaching Assistant**, Department of Computer Science, Dartmouth College
Served as the teaching assistant for COSC 74/274: Machine Learning. Gave homework tutorials and held office hours.
- Sep. 2023 – Dec. 2023  **Section Leader**, Center for Data Science, New York University
Served as the section leader for DS-GA 1018: Probabilistic Time Series Analysis. Led lab sections and held office hours.

Service

-  Reviewer: NeurIPS 2025, 2026; ICML 2026; IEEE TNNLS, TKDD

Awards

- 2024  **Dartmouth Graduate Fellowship**, merit-based.
- 2020  **Meritorious winner of Mathematical Contest in Modeling**, top 6% in 13749 teams.
- 2019-2020  **Xin-Shan Academic Excellence Award**, top 10% undergraduates university-wide.
- 2018-2021  **Academic Excellence Award**, Renmin University of China, 3x recipient.